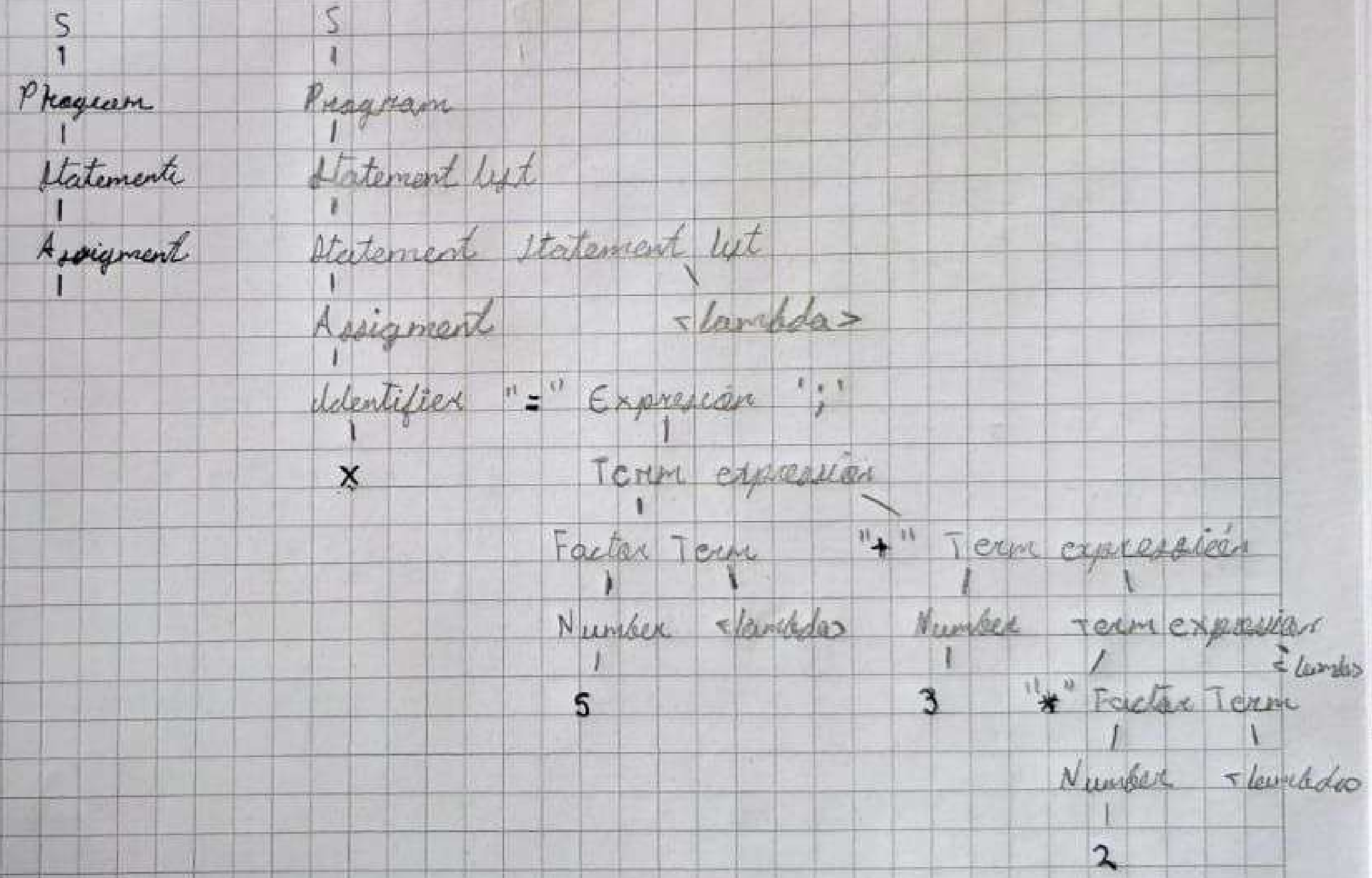
Jason Solarte 20221020153

Punto 1

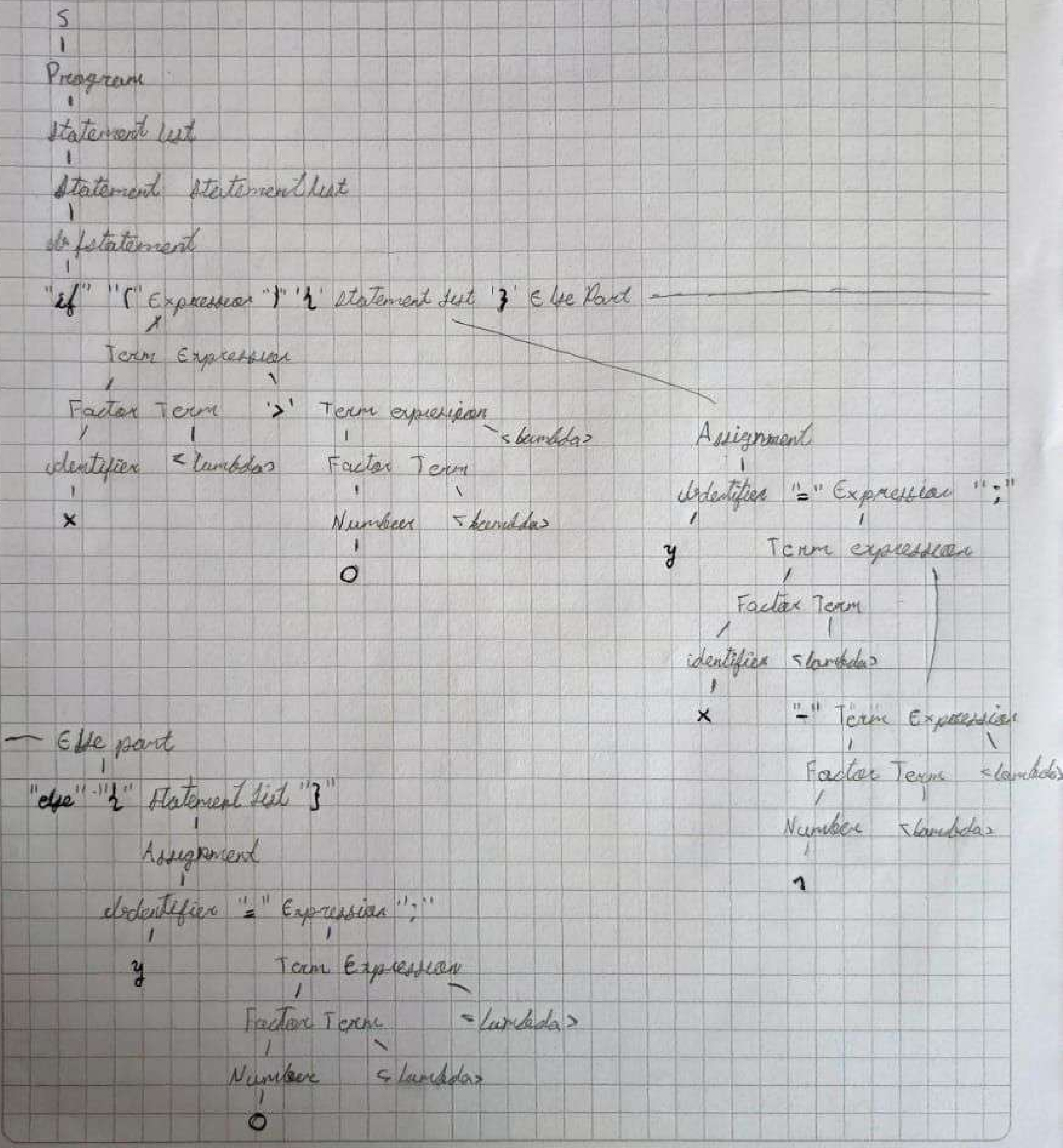
```
TOKENS = [
(IDENTIFIER'; r'[a-zA-Z][a-zA-Z0-9]*'
Integer literal r'\b\d+\b'
floating point r'\b\d+\.\d+([eE][-+]?\d+)?\b|\b\d+([eE][-+]?\d+)\b'
(string literal r'"([^"\\]|\\.[^"\\]]*)"')
(white space r'\s+')
(operator r'[+\-*/=]'
keyword r'\b(if|else|while|return)\b'
hexadecimal r'\b0[xX][0-9a-fA-F]+\b'
semicolon r';'
```

Punto 2

a. x = 5 + 3 * 2;



6.
```
if (x>0) {
   y= x-1;
} else {
   y=0;
}
```



c. while (x < 10) {

x= x + 1;

}

S

Program

Statement List

Statement Statement List — <lambda>

While Statement

while "(" Expression ")" "{" Statement List "}"

Term Expression

Factor Term "<" Term Expression

Identifier <lambda> Factor Term <lambda>

x Number <lambda>

10

— Statement List

Assignment "=" Expression ";"

x Term Expression

Factor Term "+" Term Expression

Identifier <lambda> Factor Term <lambda>

x Number <lambda>

1

4. return (a+b) * c;

S

Program

Statement List

Statement Statement List -> <lambda>

Return ~~st~~ statement

'return' Expression ';'

Term expression

Factor Term <lambda>

"(" Expression ")" "*" Factor Term <lambda>

Term expression Identifier

Factor Term "+" Term expres c

Identifier <lambda> <lambda>

a Factor Term

Identifier <lambda>

b